

Lecture 8- Part III

Market Demand

ECON101
Fall 2020



From individual to aggregate demand

Suppose the individual demand curve is given by $X_i(p)$ for individual i (suppose other determinants are held fix, for example income and prices of other goods)

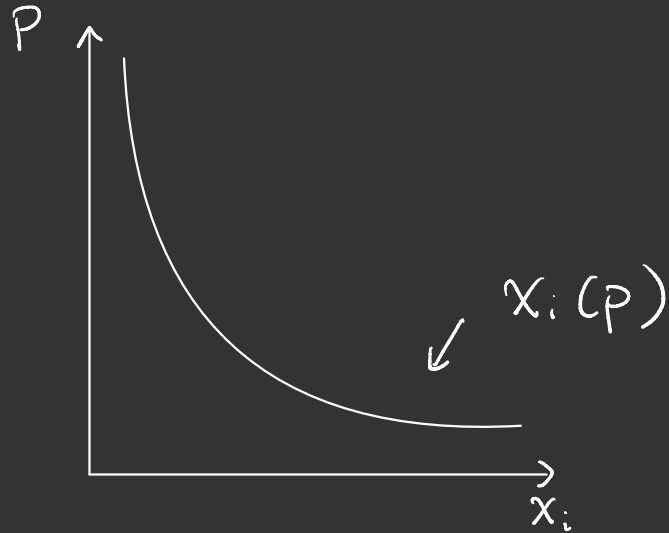
↓ Individual
Demand is only defined
for prices such that $X_i(p) \geq 0$

a convenient way to write this condition is that the individual demand curve has the expression:

$$\max \{ X_i(p), 0 \} \Leftrightarrow X_i(p) \geq 0$$

$$x_i(p) = \frac{0.3 I}{p} \xrightarrow{\text{Inverse demand}} p = \frac{0.3 I}{x_i}$$

Since for every price this demand is positive (strictly positive) there are no issues: all $p \geq 0$ work for this demand



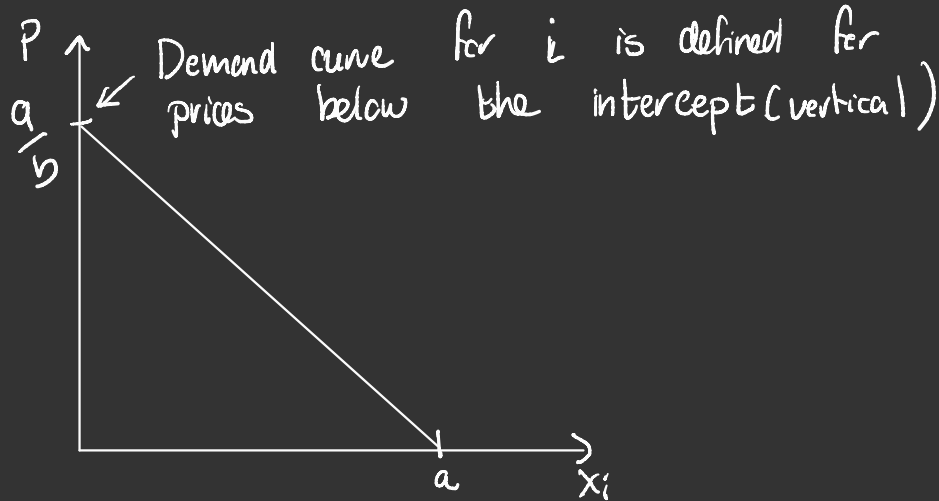
$$x_i(p) > 0 \text{ for all } p \geq 0$$

$X_i(p) = a - bp$, i.e., a linear demand curve

Define the individual demand as: $\max \{a - bp, 0\} = ?$

I need to determine when $a - bp \geq 0 \Leftrightarrow \frac{a}{b} \geq p$ Thus this demand is only defined for prices $0 \leq p \leq \frac{a}{b}$.

→ Inverse demand $p = \frac{a}{b} - \frac{1}{b} \cdot X_i$



Market Demand (for good x)

Suppose we have the demands for $i=1, \dots, n$ individuals:

$$x^1(p), x^2(p), x^3(p), \dots, x^n(p)$$

Market demand is defined as the Horizontal sum of each individual demand: for each price for which the individual demands are defined, you add them up.

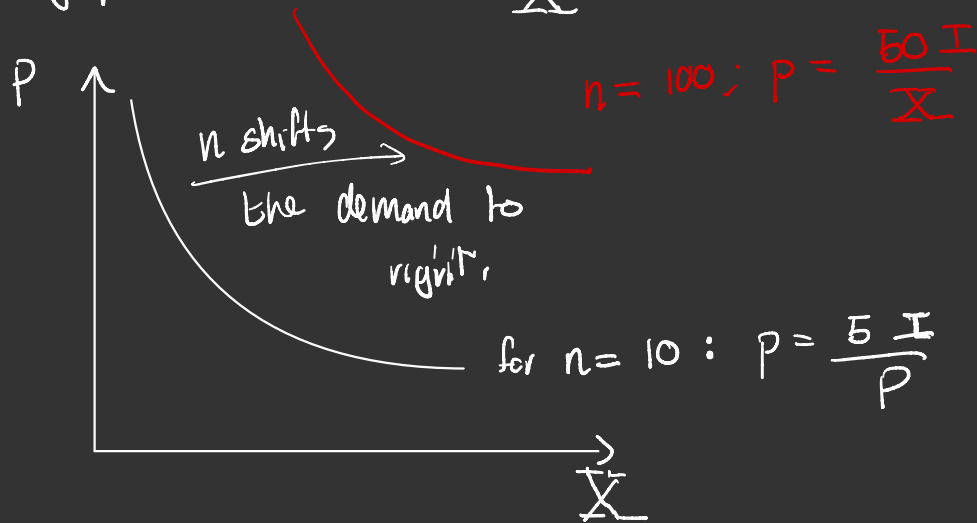
$$X(p) = \max\{x^1(p), 0\} + \max\{x^2(p), 0\} + \dots + \max\{x^n(p), 0\}$$

↑
If at a price, say $\bar{p}$, one of the demands evaluates to a negative quantity, the $\max\{\}$ takes care of it.

Example:

Suppose the individual demands are $x^i(p) = \frac{0.5 I}{p}$, $i=1, \dots, n$
all these are $x^i(p) > 0$ for all prices $p > 0$ so that
market demand is just $\bar{X}(p) = n \left(\frac{0.5 I}{p} \right) = \frac{n \times 0.5 \times I}{p}$

To graph it: $p = \frac{n \times 0.5 \times I}{\bar{X}}$



Example: Identical Linear demands:

$x^i(p) = a - bp$ for $i=1, \dots, n$, all demands are identical
all defined for $0 \leq p \leq \frac{a}{b}$, the market demand is:

$$X(p) = n(a - bp) = na - nbp$$

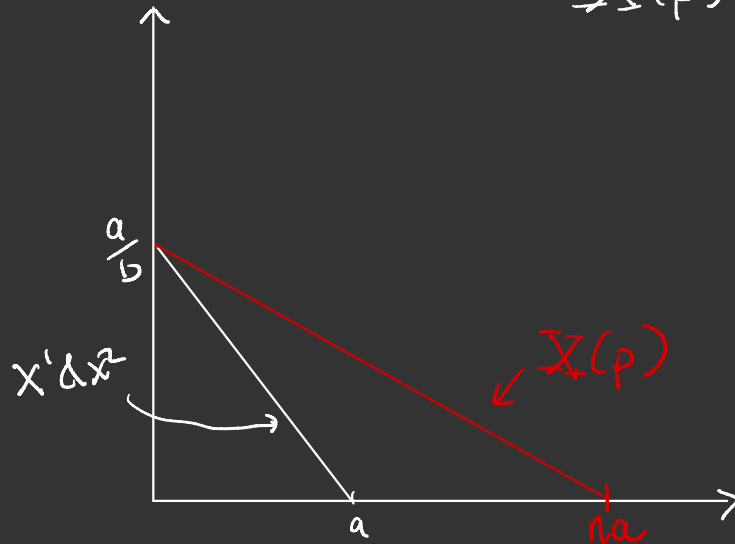
$$p = \frac{a}{b} - \frac{x}{nb}$$

$$X(p) = na - nbp \text{ for } 0 \leq p \leq \frac{a}{b}$$

Suppose $n=2$

$$x^1(p) = a - bp$$

$$x^2(p) = a - bp$$



Example:

$$x^1(p) = 10 - p; \quad x^2 = 20 - 2p;$$

Inverse demands:

$$p = 10 - x^1$$
$$p = 10 - \frac{x^2}{2}$$

$$X(p) = \max\{10 - p, 0\} + \max\{20 - 2p, 0\} = 30 - 3p \quad \text{for } 0 \leq p \leq 10$$

$$x^1(p) \geq 0 \quad \text{if } p \leq 10$$

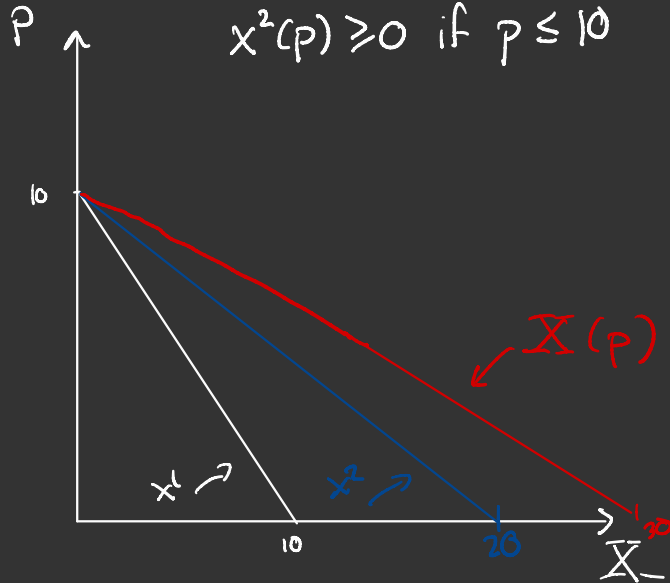
$$x^2(p) \geq 0 \quad \text{if } p \leq 10$$

add up

$$x^1(p) + x^2(p) \quad \text{for prices } p \leq 10$$

$$X(p) = 30 - 3p \quad \text{for } 0 \leq p \leq 10$$

$$p = 10 - \frac{X}{3}$$



Example:

$$Q^{DD}(P) = 1430 - 55P \Rightarrow Q^{\text{world}}(P) = \max \{1430 - 55P, 0\} + \max \{1470 - 70P, 0\}$$
$$Q^{DE}(P) = 1470 - 70P$$

$$Q^{DD}(P) \geq 0 \Leftrightarrow 1430 - 55P \geq 0 \Leftrightarrow \frac{1430}{55} = 26 \geq P$$

$$Q^{DE}(P) \geq 0 \Leftrightarrow 1470 - 70P \geq 0 \Leftrightarrow \frac{1470}{70} = 21 \geq P$$

$$Q^{\text{world}}(P) = \begin{cases} 1430 - 55P & \text{for } 21 < P \leq 26 \\ 2900 - 125P & \text{for } P \leq 21 \end{cases}$$

$$Q^{DD}(P) = 1430 - 55P \Rightarrow Q^{\text{WORLD}}(P) = \max\{1430 - 55P, 0\} + \max\{1470 - 70P, 0\}$$

$$Q^{DE}(P) = 1470 - 70P$$

$$Q^{DD}(P) \geq 0 \Leftrightarrow 1430 - 55P \geq 0 \Leftrightarrow \frac{1430}{55} = 26 \geq P$$

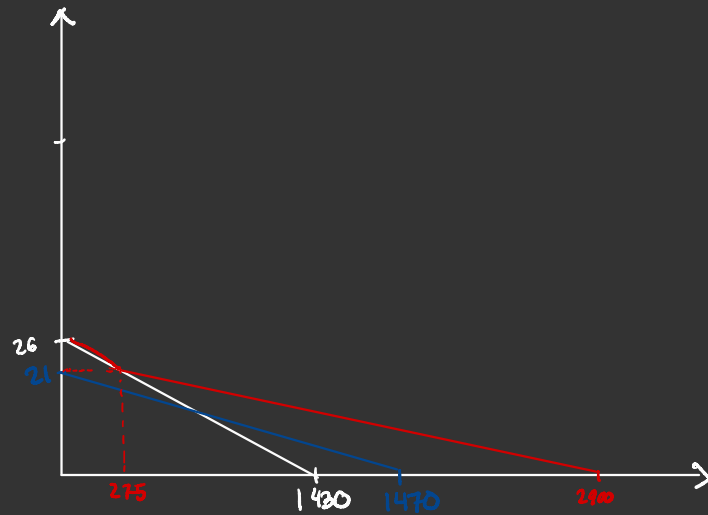
$$Q^{DE}(P) \geq 0 \Leftrightarrow 1470 - 70P \geq 0 \Leftrightarrow \frac{1470}{70} = 21 \geq P$$

$$Q^{\text{WORLD}}(P) = \begin{cases} 1430 - 55P & \text{for } 21 < P \leq 26 \\ 2900 - 125P & \text{for } P \leq 21 \end{cases}$$

For Q^{DD} : $P = 26 - \frac{1}{55} Q^{DD}$

For Q^{DE} : $P = 21 - \frac{1}{70} Q^{DE}$

For Q^{WORLD} : $P = \begin{cases} 26 - \frac{1}{55} Q^{\text{WORLD}} \\ 23.2 - \frac{1}{125} Q^{\text{WORLD}} \end{cases}$



$$0 \leq Q^{\text{WORLD}} < 1430 - 55(21) = 275$$

$$2900 - 125(21) = 275 \leq Q^{\text{WORLD}} \leq 2900$$